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Drive through system including vision system and transaction system integration

Abstract

A control system for a drive through includes processing circuitry. The processing circuitry is configured to determine, using image data obtained from multiple cameras, a characteristic of a vehicle in a lane of the drive through. The processing circuitry can operate a point of sale unit to provide an indication of the characteristic of the vehicle. The processing circuitry can, responsive to detection of one or more of a plurality of vehicles using the image data, perform at least one of (i) initiating an alert via the point of sale unit for staff of the drive through, (ii) initiating an action to open an additional lane, or (iii) initiating a kitchen action to prepare one or more food or beverage items.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application claims the benefit of and priority to (i) U.S. Provisional Application No. 63/452,218, filed Mar. 15, 2023, (ii) U.S. Provisional Application No. 63/539,920, filed Sep. 22, 2023, and (iii) U.S. Provisional Application No. 63/587,611, filed Oct. 3, 2023, the disclosures of each of which are incorporated by reference herein in their entireties.

FIELD

(1) The present disclosure relates to vision systems. More specifically, the present disclosure relates to vision systems for drive throughs.

BACKGROUND

(2) Drive through systems may have multiple lanes with a combination of display interfaces, microphones, speakers, and vehicle detection capabilities. When a customer arrives at the drive through system, the customer may communicate via a menu board or unit with an order taker to place their order. The customer then pulls around to pay and pick up the customer's order. Cameras may obtain image data of vehicles at the drive through system.

SUMMARY

(3) One implementation of the present disclosure is a drive through system, according to some embodiments. The drive through system can include a transaction system, such as a point of sale unit, configured to be operated to implement a point of sale. The drive through system can also include multiple cameras configured to obtain image data of a lane of the drive through system. The drive through system can also include processing circuitry configured to determine, using the image data, a characteristic of a vehicle at the lane, and operate the point of sale unit to provide an indication of the characteristic of the vehicle. The processing circuitry may also be configured to, responsive to detection of one or more of multiple vehicles using the image data, perform at least one of (i) initiating an alert via the point of sale unit for staff of the drive through system, (ii) initiating an action to open an additional lane, or (iii) initiating a kitchen action to prepare one or more food or beverage items.

(4) The characteristic of the vehicle may include a type of the vehicle selected from multiple predetermined types of vehicles using the image data, and a color of the vehicle. The multiple predetermined types of vehicles can include at least a car, a truck, and an SUV.

(5) The processing circuitry may be further configured to determine, using the image data from one of multiple cameras or the multiple cameras and predetermined zones, a location of the vehicle along a path of the lane. The predetermined zones may be defined for the image data of each of the multiple cameras to account for different orientations and positions of the cameras that results in image data of the predetermined zones spanning different portions of the image data of each of the cameras. The processing circuitry can also be configured to, responsive to the vehicle arriving at an order placement location of the zone, operating the point of sale unit to provide the indication of the characteristic of the vehicle.

(6) The indication of the characteristic of the vehicle can include at least one of a textual indication of the characteristic of the vehicle or a visual representation of the characteristic of the vehicle. The drive through system can also include at least one of a point of sale unit or a display screen (e.g., notifier, alarm) at a pickup window of the drive through system viewable by an employee of the drive through system. The processing circuitry may be configured to operate at least one of the point of sale unit or the display screen at the pickup window to provide order data associated with the vehicle and the characteristic of the vehicle responsive to the vehicle arriving at the pickup window.

(7) The processing circuitry can be configured to track multiple vehicles through the lane the drive through system using the image data obtained from the cameras. The processing circuitry can also be configured to, responsive to a number of the plurality of vehicles, at least one of initiate an alert for staff of the drive through system, initiate an action to open an additional lane, or initiate a

kitchen action to prepare a quantity of food items. The processing circuitry can also be configured to operate, based on the image data, a carry-out alert device to notify carry-out staff regarding (i) the characteristic of a vehicle tracked to a carry-out location, (ii) an identification of which carry-out location of a plurality of carry-out locations at which the vehicle is positioned, and (iii) order data for the vehicle at the carry-out location.

(8) Another implementation of the present disclosure is a method, according to some embodiments. The method may include obtaining image data of a lane of a drive through system and determining, using the image data, a visual characteristic of a vehicle at the lane. The method can also include operating a point of sale unit to provide an indication of the visual characteristic of the vehicle. The point of sale unit may be configured to be operated to implement a point of sale. The visual characteristic may include a type of the vehicle selected from multiple predetermined types using the image data, and a color of the vehicle. The method can also include, processing, at the point of sale unit, vehicle information relating to the visual characteristic of the vehicle and order information for an order received from an occupant of a drive through to provide compiled order information. The method may also include updating multiple display devices of the drive through system to include an indication of the compiled order information during a service journey of the vehicle at the drive through.

(9) The multiple types of vehicles can include at least a car, a truck, and an SUV. The method can also include determining, using the image data and multiple predetermined zones, a location of the vehicle along a path of the lane. The method can also include, responsive to the vehicle arriving at an order placement location of the zone, operating multiple point of sale units and at least one display screen to provide the indication of the characteristic of the vehicle to staff of the drive through system. The multiple point of sale units may be positioned at different employee stations of the drive through system and the display screen may be positioned in a kitchen or at a pickup window of the drive through system.

(10) The indication of the visual characteristic of the vehicle can include at least one of a textual indication of the characteristic of the vehicle or a visual representation of the visual characteristic of the vehicle. The method can further include operating an alert device to provide an indication of the compiled order information responsive to the vehicle arriving at a pickup window. The alert device may be positioned at a pickup window of the drive through system and viewable by an employee of the drive through system.

(11) The method can also include tracking multiple vehicles through the lane the drive through system using the image data obtained from multiple cameras. The method can also include, responsive to a number of the vehicles, at least one of initiating an alert for staff of the drive through system, initiating an action to open an additional lane, or initiating a kitchen action to prepare a quantity of food items. The method can also include operating a carry-out alert device to provide an indication of the compiled order information responsive to the vehicle arriving at a carry-out location. The carry-out alert device can also be configured to provide an indication of which of multiple carry-out locations at which the vehicle is located.

(12) Another implementation of the present disclosure is a control system for a drive through, according to some embodiments. The control system includes processing circuitry, according to some embodiments. The processing circuitry may determine, using image data obtained from multiple cameras, a characteristic of a vehicle in a lane of the drive through. The processing circuitry can operate a point of sale unit to provide an indication of the characteristic of the vehicle. The processing circuitry can, responsive to detection of one or more of a plurality of vehicles using the image data, perform at least one of (i) initiating an alert via the point of sale unit for staff of the drive through, (ii) initiating an action to open an additional lane, or (iii) initiating a kitchen action to prepare one or more food or beverage items.

(13) In some embodiments, the processing circuitry is further configured to, responsive to detecting that a vehicle has arrived at an order fulfillment location, operate a display screen at the order

fulfillment location to display an initially obtained image of the vehicle to an order fulfillment staff to facilitate accurate order fulfillment. In some embodiments, the point of sale unit is a first point of sale unit of multiple point of sale units. The processing circuitry is configured to operate the point of sale units to provide the indication of the characteristic of the vehicle, and a location of the vehicle in the drive through on display screens of the point of sale units in unison, according to some embodiments. In some embodiments, the point of sale units are each configured to receive a user input to transition a corresponding one of multiple portable communications devices to communicate on an audio channel according to the user input. In some embodiments, the processing circuitry is further configured to operate a kitchen display screen to provide order information, the characteristic of the vehicle, and a location of the vehicle to kitchen staff of the drive through.

(14) Another implementation of the present disclosure is a point of sale system for a drive through, according to some embodiments. The point of sale system can include a point of sale unit and processing circuitry. The point of sale unit can be configured to be operated to implement a point of sale and obtain order data. The processing circuitry can be configured to determine, using image data obtained from a camera, a characteristic of a vehicle at a lane of the drive through. The processing circuitry may also be configured to operate the point of sale unit to provide an indication of the characteristic of the vehicle.

(15) The characteristic of the vehicle include a type of the vehicle selected from multiple predetermined types of vehicles using the image data, and a color of the vehicle. The multiple predetermined types of vehicles can include at least a car, a truck, and an SUV.

(16) The processing circuitry can further be configured to determine, using the image data and multiple predetermined zones, a location of the vehicle along a path of the lane. The processing circuitry can also be configured to, responsive to the vehicle arriving at an order placement location of the zone, operate the point of sale unit to provide the indication of the characteristic of the vehicle.

(17) The indication of the characteristic of the vehicle can include at least one of a textual indication of the characteristic of the vehicle or a visual representation of the characteristic of the vehicle. The point of sale system can include an alert device at a pickup window of the drive through system viewable by an employee of the drive through system. The processing circuitry may be configured to operate the alert device to provide the order data associated with the vehicle and the characteristic of the vehicle responsive to the vehicle arriving at the pickup window.

(18) The processing circuitry can be configured to track multiple vehicles through the lane the drive through system using the image data obtained from multiple cameras. The processing circuitry can also, responsive to a number of the multiple vehicles, at least one of initiate an alert for staff of the drive through system, initiate an action to open an additional lane, or initiate a kitchen action to prepare a quantity of food items.

(19) This summary is illustrative only and is not intended to be limiting. Various aspects, inventive features, and advantages of the systems described herein are set forth in more detail below.

(20) Various aspects of the present disclosure, such as the graphical user interfaces (GUIs) shown and described, can be implemented in connection with the drive-through systems and interfaces set forth in U.S. Provisional Application No. 63/539,920 filed Sep. 22, 2023, which is incorporated herein by reference in its entirety.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying FIGURES, wherein like reference numerals refer to

like elements, in which:

- (2) FIG. 1 is a diagram of a drive through system including four lanes and a communications system for communicating with customers of the four lanes, according to some embodiments.
- (3) FIG. 2 is a diagram of an order station of the drive through system of FIG. 1 at which a customer may place an order by speaking or causing an audible input to be provided into a microphone of a drive through unit that is in communications with an order taker of a restaurant, according to some embodiments.
- (4) FIG. 3 is a block diagram of a setup system for determining a setup profile of a vision system that is integrated with a point of sale system of a store, according to some embodiments.
- (5) FIG. 4 is a flow diagram of a process for operating point of sale units of a point of sale system using image data of a drive through system, according to some embodiments.
- (6) FIG. 5 is satellite data of a store with a defined drive through route and camera locations, according to some embodiments.
- (7) FIG. 6 is a graphical user interface (GUI) of a user interface of the setup system of FIG. 3, according to some embodiments.
- (8) FIG. 7 is another GUI of the user interface of the setup system of FIG. 3, according to some embodiments.
- (9) FIG. 8 is a diagram of a setup profile of a drive through system including different lanes, according to some embodiments.
- (10) FIG. 9 is a block diagram of a point of sale system for the drive through system of FIG. 1 that integrates a vision system with point of sale units, according to some embodiments.
- (11) FIG. 10 is a block diagram of a controller of the point of sale system of FIG. 9, according to some embodiments.
- (12) FIG. 11 is a flow diagram of a process for operating point of sale units of a point of sale system of a drive through using image data from a vision system, according to some embodiments.
- (13) FIG. 12 is a GUI of a point of sale unit that is updated using image data of a vision system, according to some embodiments.
- (14) FIG. 13 is another GUI of a point of sale unit that is updated using image data of the vision system, according to some embodiments.
- (15) FIG. 14 is another GUI of a point of sale unit that is updated using image data of the vision system, according to some embodiments.
- (16) FIG. 15 is another GUI of a point of sale unit that is updated using image data of the vision system, according to some embodiments.
- (17) FIG. 16 is a GUI illustrating tracking of vehicles in a drive through system, according to some embodiments.
- (18) FIG. 17 is another GUI illustrating tracking of vehicles in a drive through system including a stack size, according to some embodiments.
- (19) FIG. 18 is another GUI illustrating tracking of vehicles in a drive through system including a leaderboard, according to some embodiments.
- (20) FIGS. 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 and 31 are exemplary GUIs that may be presented on a point of sale unit or a director display screen of a drive through system, according to some embodiments.
- (21) FIG. 32 is a GUI that may be presented on a kitchen display screen of a drive through system, according to some embodiments.

DETAILED DESCRIPTION

(22) Before turning to the FIGURES, which illustrate the exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the FIGURES. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

(23) Overview

(24) Referring generally to the FIGURES, a vision system for a drive through system can include cameras that track vehicles through their life cycle of the drive through system. The vision system may implement a machine learning technique in order to automatically identify a type and color of the vehicles in the drive through system. The vision system is integrated with a transaction system, e.g., a system of point of sale units, such that the point of sale units are updated to indicate the type and color of the vehicles. Integrating the vision system with the point of sale units facilitates improved speed and order fulfillment accuracy of the drive through system.

(25) Drive Through System

(26) Referring particularly to FIGS. 1-2, a drive through system **10** for a store **14** (e.g., a restaurant, a diner, a fast food restaurant, an establishment, etc.) includes a lane system **16** (e.g., a starting lane, one or more ordering lanes where a transaction such as a point of sale (POS) transaction may be conducted or initiated, ordering lanes where customers may communicate with employees of the store **14**, etc.), a starting location **32**, and an ending location **34**. The starting location **32** may be a zone or area at which one or more customers (e.g., customers in vehicles, shown as vehicle **30**) enter the lane system **16**. The lane system **16** includes an entry lane **18**, a first order lane **20a**, a second order lane **20b**, a third order lane **20c**, a fourth order lane **20d**, and a pickup lane **22**. The pickup lane **22** (e.g., an exit lane) may extend along a side of the store **14** including one or more pickup windows, shown as first window **28a**, second window **28b**, and third window **28c** such that customers (e.g., vehicles **30**) may access the different windows **28** to pick up their orders from the store **14**. The term “window” should be interpreted broadly to mean a customer-store interface which is not limited to those with a physical screen or barrier.

(27) The starting location **32** and the ending location **34** may define starting and ending points for customers, with the order lanes **20** defining different paths between the starting location **32** and the ending location **34**. In some embodiments, each of the order lanes **20** defines a corresponding order zone **24** at which the customer may place an order for pickup at one of the windows **28**. In some embodiments, the first lane **20a** includes a first order zone **24a**, the second lane **20b** includes a second order zone **24b**, the third lane **20c** includes a third order zone **24c**, and the fourth lane **20d** includes a fourth order zone **24d**. Each of the order zones **24** includes a corresponding drive through unit **108** (e.g., a menu board, a kiosk, a field communications unit, an order placement unit at which a customer may place an order, etc.) which facilitates communications between the customer at the order zones **24** and personnel of the store **14**, as well a display screen or signage indicating available items for purchase, and order or POS information (e.g., a number of items, corresponding cost, total cost, tax, etc., for an ongoing order). In some embodiments, the first order lane **20a** includes a first drive through unit **108a**, the second order lane **20b** includes a second drive through unit **108b**, the third order lane **20c** includes a third drive through unit **108c**, and the fourth order lane **20d** includes a fourth drive through unit **108d**. In some embodiments, the order lanes **20** define one or more paths that include a series of discrete geometric forms (e.g., polygonal shapes, curvilinear shapes) mapped to different physical locations along the order lanes **20**. The series of geometric forms may match between multiple cameras (e.g., cameras **114**) that have different field of views in order to facilitate awareness, identification, and tracking of vehicles along the one or more paths between the starting location **32** and the ending location **34**.

(28) The drive through units **108** are interfaces that may be components of a communications system or POS system, shown as communications system **100** of the drive through system **10**. In some embodiments, the drive through units **108** are integrated via one or more POS systems. The communications system **100** includes the drive through units **108**, a controller **102**, one or more imaging devices **114** (e.g., cameras) positioned about the lane system **16**, a cloud computing system **104**, one or more input devices, shown as order taking stations **110**, and one or more wearable communications devices **112**. In some embodiments, the one or more order taking stations **110** are disposed in each of a corresponding zone within the store **14** proximate the windows **28**. The order

taking stations **110** may include a touch screen or user interface configured to both display image data (e.g., a graphical user interface, a menu, selectable options for an order or POS, etc.), and receive user inputs from a corresponding employee of the store **14** to add items to an order or POS. (29) The wearable communications devices **112** may have the form of headphones, earpieces, etc., and can include both speakers (e.g., acoustic transducers, aural output devices, etc.) and microphones (e.g., aural input devices). In some embodiments, the controller **102** is communicably coupled with each of the order taking stations **110** at the windows **28**, the imaging devices **114**, and the head wearable communications devices **112**. The controller **102** may also be communicably coupled with a cloud computing system **104** and can upload or provide various information regarding points of sale to the cloud computing system **104** for analysis. In some embodiments, the controller **102** is configured to receive, from the cloud computing system **104**, software or firmware updates for various devices of the communications system **100**. In some embodiments, the wearable communications devices **112** may be portable communications devices including but not limited to headphones, earphones, earbuds, devices configured to clip to a belt or article of clothing, ear pieces, etc., any device having at least one of a speaker or microphone and configured to be worn, held, or otherwise move with an individual to establish an end point for audio communications (e.g., to at least one of produce sound via a speaker, or receive a sound input via a microphone). It should be understood that the components of the communications system **100** (e.g., the drive through units **108**, the controller, the one or more imaging devices **114**, the cloud computing system **104**, the order taking stations **110**, and the wearable communications devices **112**) may optionally be integrated with each other via a POS (e.g., all such components or a subset of the foregoing components). Integrating the components of the communications system **100** via the POS may facilitate improved efficiency of the drive through, for example.

(30) The communications system **100** provides selective communicability according to multiple channels between any of the personnel (e.g., personnel **40a** at the first window **28a**, personnel **40b** at the second window **28b**, and personnel **30c** at the third window **28c**) and customers at one of the order zones (e.g., the first order zone **24a**, the second order zone **24b**, the third order zone **24c**, and the fourth order zone **24d**) via the wearable communications devices **112** and the drive through units **108**. In some embodiments, each of the drive through units **108** is configured to communicate according to a corresponding channel (e.g., a first order channel, a second order channel, a third order channel, and a fourth order channel) and the personnel at the windows **28** can selectively establish communicability with the customers at the different order lanes **20** by selectively switching between the channels such that the wearable communications devices **112** are communicatively coupled on a selected one of the channels. In some embodiments, the communications system **100** also facilitates private communication between one or more of the personnel **40** within the store **14** or at other remote locations (e.g., a service center). For example, the personnel may include kitchen personnel **40d** that may communicate with any of the personnel **40a**, the personnel **40b**, or the personnel **40c** at the windows **28** via one or more remote channels. In some embodiments, the kitchen personnel **40d** may also listen on one or more of the channels on which the customers at the zones **24** can communicate.

(31) Referring particularly to FIG. 2, one of the order zones **24** is shown in greater detail, according to some embodiments. The order zones **24** may include a space or surface of the order lane **20** for the customer, shown as vehicle **30** to pull up and place an order. The order zone **24** may include at least one pressure or force sensor **106**. The at least one pressure or force sensor **106** is configured to detect the presence of the vehicle **30** and notify the controller **102** that a customer has arrived to the specific order zone **24** (e.g., pulled up, reversed into, merged into, parked in, etc.). The order zones **24** may also include one or more imaging devices **114** (e.g., cameras, image sensors, sensor arrays) for detecting the presence of the vehicle **30**. The drive through units **108** include a display screen **118** configured to display a current order, items in the order, and cost of the order or other aspects of an order property, alone or in any combination. The display screen **118** may be operable by one

or more of the personnel **40** who is taking the customer's order. In some embodiments, the order zones **24** may optionally include at least one pressure sensor and/or at least one magnetic ground loop. For example, if the order zones **24** already include a pressure sensor and/or a magnetic ground loop, the controller **102** may communicate with and use feedback from the pressure sensor and/or the magnetic ground loop. In some embodiments, the order zones **24** may lack both a pressure sensor and/or a magnetic ground loop.

(32) The drive through unit **108** may also include a microphone **122** configured to capture audio (e.g., spoken audio) from the customer and transmit data (audio data, audio signals) to the personnel **40** that is taking the customer's order via a corresponding channel of the communications system **100** for the order lane **20** of the order zone **24**. In some embodiments, the drive through unit **108** also includes a speaker **120** configured to provide audio data of the voice or spoken words of the personnel **40** obtained by the wearable communication device **112** that is communicating on the channel of the drive through unit **108**. In this way, the communications unit **108** may provide a customer end for audio communications and the wearable communications device **112** may provide a personnel end for audio communications between the customer and one of the personnel **40** (e.g., according to an audio or communications channel). In some embodiments, the at least one pressure or force sensor **106**, the speaker **120**, the microphone **122**, the imaging device **114**, and the display screen **118** are configured to communicate directly with the controller **102** of the communication system **100**. In some embodiments, the pressure or force sensor **106**, the speaker **120**, the microphone **122**, the imaging device **114**, and the display screen **118** are configured to communicate with a zone controller **124** that communicates with the controller **102**.

(33) Referring again to FIG. **1**, the controller **102** and the imaging devices **114** may form a vision system **200**. The vision system **200** can be implemented on the controller **102** based on image data received from the imaging devices **114**. In some embodiments, the vision system **200** is configured to integrate with one or more POS units of the communications system **100**, such as the order taking units **110**. The vision system **200** is configured to track vehicles or customers through the drive through system **10** (e.g., along different lanes) and integrate with a POS system. The vision system **200** may trigger notifications to the personnel **40**. In some embodiments, the vision system **200** is configured to notify the personnel **40** regarding the presence of a vehicle at one of the drive through units **108**. In some embodiments, the vision system **200** is configured to identify which vehicle has arrived at a pickup window and notify associated personnel **40** regarding which order the vehicle at the pickup window should receive. In some embodiments, the vision system **200** is configured to prompt the personnel **40** to open another order taking or drive through lane. In some embodiments, the vision system **200** is configured to notify the kitchen personnel **40d** regarding an influx of vehicles **30** to the drive through system **10** to prompt the kitchen personnel **40d** to begin preparing food in advance.

(34) Advantageously, the vision system **200** may track vehicles through the drive through system **10** and integrate with POS systems of the drive through system **10**. The vision system **200** may facilitate improved order taking, speed of service, and fulfillment accuracy. The vision system **200** integrates directly with the POS units such that graphical user interfaces of the POS units may be updated in real-time based on tracking of vehicles **30** through the drive through system **10** by the vision system **200**.

(35) Vision System

(36) Offline Setup

(37) Referring to FIG. **3**, a setup system **300** for configuring the vision system **200** includes a setup device **302**, a user interface **304**, the controller **102**, the cloud computing system **104**, and a satellite **306**, according to some embodiments. The setup device **302** is configured to facilitate setting up and configuring the controller **102** to perform image analysis techniques and tracking of vehicles **30** through the drive through system **10**. In some embodiments, the setup device **302** is configured to generate a profile or configuration settings for the controller **102**. In some embodiments, the

setup device **302** is configured to determine the profile or configuration settings for the controller **102** for a specific location or drive through system **10**. For example, the setup device **302** may implement one or more set-up functions (e.g., a guide for a user) to determine the profile or configuration settings for the controller **102**. The profile or the configuration settings may include paths corresponding to different lanes of traffic of the drive through system **10**, and one or more geometric shapes or zones along the paths. The profile or the configuration settings may be location-specific and tailored to the lanes of the specific location of the drive through system **10**. In some embodiments, the profile or the configuration settings include one or more preferred operating or alert settings for the store **14** (e.g., for kitchen personnel, for order takers, etc.). In some embodiments, the profile or the configuration settings include integration for the tracking of the vehicles **30** with one or more POS units. It should be understood that any of the functionality of the setup device **302** may be performed locally on the controller **102**, or remotely on the cloud computing system **104**.

(38) The setup device **302** may obtain satellite image data of the drive through system **10** from the satellite **306**. The satellite image data may indicate different lanes of traffic, emergency lanes, parking lots, etc., of a store or restaurant. The setup device **302** is configured to operate the user interface **304** in order to display the satellite image data to a user, according to some embodiments. In some embodiments, the setup device **302** is configured to operate the user interface **304** to prompt the user to provide inputs indicative of one or more paths of the drive through system **10**. The paths of the drive through system **10** may correspond to different lanes of traffic for the store **14**. The user may provide to the setup device **302**, via the user interface **304**, one or more points indicating locations along a first path. The user may provide points corresponding to multiple different paths (e.g., a second path, a third path, etc.), corresponding to a number of lanes of traffic or routes. In some embodiments, the setup device **302** is configured to receive the points for each of the one or more paths and perform a curve fitting technique in order to generate the path based on the points. For example, the setup device **302** may implement a variety of curve fitting techniques based on the points, including but not limited to Bezier curves, B-spline curves, Catmull-Rom splines, interpolating Lagrange curves, etc. In some embodiments, the setup device **302** uses the point to generate paths or curves including control points. The control points may correspond to the points provided by the user via the user interface **304** that the setup device **302** uses to generate the curves or paths. In some embodiments, the points provided to the setup device **302** are global positioning system (GPS) coordinates such as latitude and longitude which are used by the setup device **302** to define the curves.

(39) Once the paths or curves for each lane of traffic are determined by the setup device **302**, the setup device **302** may operate the user interface **304** to display the paths or curves. The setup device **302** may also receive updates to the curves or paths as user inputs from the user interface **304**. The updates to the curves or paths (e.g., the Catmull-Rom splines) may include the addition of or subtraction of control points, the adjustment (e.g., movement) of control points, changing an amount or direction of a tangent of one or more of the control points, etc.

(40) The curves may generally define the lanes of traffic of vehicles **30** through the drive through system **10** and facilitate tracking the vehicles **30** through the drive through system **10** (e.g., from starting location **32** to the ending location **34**). In the example drive through system shown in FIG. **1**, six different curves may be defined by the user: (i) a first path or curve defined along the entry lane **18**, (ii) a second path or curve defined along the first order lane **20a**, (iii) a third path or curve defined along the second order lane **20b**, (iv) a fourth path or curve defined along the third order lane **20c**, (v) a fifth path or curve defined along the fourth order lane **20d**, and (vi) a sixth path or curve defined along the pickup lane **22**.

(41) In some embodiments, the setup device **302** is configured to determine, based on the paths or curves defined by the user for the drive through system **10**, one or more suggested locations for the cameras **114**. If the cameras **114** are already installed such that the cameras **114** can obtain image

data from a variety of locations of all of the paths or curves, the setup device **302** may proceed to generation and customization of different tracking zones along the paths or curves. In some embodiments, the setup device **302** is configured to select from a variety of different cameras and determine, based on the cameras fields of view, positions and orientations for multiple cameras such that image data of all of the paths or curves can be obtained. In some embodiments, the setup device **302** is configured to provide, via the cloud computing system **104**, the satellite image data and the curves or paths superimposed on the satellite image data, to a system administrator of the setup system **300**. The system administrator (e.g., a manufacturer of the vision system **200**) may provide recommended locations, orientations, and types of cameras for the drive through system **10**. The system administrator may further provide, to the setup device **302** via the cloud computing system **104**, recommended locations, orientations, and types of cameras for the drive through system **10** based on an on-site or remote inspection, according to some embodiments.

(42) The setup device **302** may be configured to determine multiple zones (e.g., geometric shapes, areas, etc.) along the paths or curves. The setup device **302** may automatically determine the zones along the paths or curves and operate the user interface **304** to display the satellite image data, the paths or curves, the points, and the zones superimposed on the satellite image data. In some embodiments, the user interface **304** is configured to use selections, provided by the user interface **304** as user inputs, and provide different image data from corresponding cameras **114** to the user via the user interface **304** in response to the selections. In some embodiments, the setup device **302** is configured to display the image data to the user interface **304** such that the user may define the zones.

(43) The user interface **304** is configured to receive user inputs to define points in the image data of each of the cameras **114** to define zones along the curves or paths. In some embodiments, the setup device **302** is configured to receive four points for each zone in the image data. The setup device **302** may receive four points for each of the zones, and allow, via operation of the user interface **304**, the user to define zones for all of the lanes of traffic that are visible in the image data of one of the cameras **114**. After the user has defined the zones for the image data of a first camera **114**, the user may select a second camera **114**, and proceed to define correspondingly numbered zones for the image data of the second camera **114**. This process may be repeated by the user via the user interface **304** and the setup device **302** until all of the zones have been defined for the image data of each of the cameras **114**.

(44) The user may be prompted to define the zones and assign corresponding numbers or labels in a uniform manner across the image data from different cameras **114**. For example, if a zone A is defined in front of a pickup window, then the zone A should be in front of the pickup window for the image data across different cameras **114**. Advantageously, the zones can be used by the controller **102** (e.g., in the profile) in order to track the vehicle **30** through the drive through system **10**.

(45) Referring still to FIG. 3, the setup device **302** may receive inputs from the user interface **304** in order to define one or more masks for the image data obtained from the cameras **114**. For example, if the first camera **114** includes image data of surrounding areas that vehicles **30** will not be detected in, the user inputs may define a mask for the surrounding areas. In some embodiments, the setup device **302** is configured to provide display data to the user interface **304** such that the user interface **304** displays the image data. The user interface **304** may provide an “eraser tool” to the user, a pen tool, a paintbrush tool, etc., such that the user can provide inputs to define the masks. The setup device **302** may store information of the masks of the image data as a part of the setup profile.

(46) The setup device **302** may also receive user inputs from the user interface **304** defining locations of interest along the paths, curves, or routes. The locations of interest may include an arrival location, a drive through entry location, an order lane entry location, an order location, an order lane exit location, a payment location, an order pickup location, an exit location, etc. In some

embodiments, the locations are stored by the setup device **302** and used in order to determine the setup profile. The locations may be point locations or may be tied to corresponding zones. For example, the locations may be labels associated with corresponding zones (e.g., zone **501** is the order pickup zone).

(47) The setup device **302** is configured to provide the setup profile to the controller **102** for use in operating the drive through system **10**, according to some embodiments. In some embodiments, the setup profile includes (i) the paths, routes, or curves for the different lanes of traffic (e.g., entry lanes, parking lot lanes or routes, order lanes, pickup lanes, emergency lanes, exit lanes, etc.), (ii) the zones along the lanes of traffic (e.g., a series of geometric shapes that follow the lanes of traffic) for the satellite image data, and the image data from each of the cameras **114**, (iii) one or more locations of interest, (iv) and an image analysis technique or set of instructions. In some embodiments, the image analysis technique is provided in the form of a neural network, an artificial intelligence, a machine learning technique or set of instructions, etc. The image analysis technique may be performed by the controller **102** locally such that the controller **102** can use the image data provided by the cameras **114** in order to detect a type and color of vehicle. In some embodiments, performing the image analysis technique locally at the controller **102** of the drive through system **10** facilitates ensuring privacy of the image data.

(48) Referring to FIG. **4**, a flow diagram of a process **400** for setting up a vision tracking system and implementing the vision tracking system in an integrated manner with a POS unit includes steps **402-418**, according to some embodiments. In some embodiments, the process **400** is implemented by the setup system **300** or components thereof, and the controller **102** of the drive through system **10**. In some embodiments, the process **400** is implemented in order to set up a profile in a location-specific area or for a specific restaurant location.

(49) The process **400** includes obtaining map data of a drive through location (step **402**), according to some embodiments. In some embodiments, the map data of the drive through location is satellite image data. In some embodiments, the map data of the drive through location is obtained by the setup device **302** from the cloud computing system **104** or the satellite **306**. The map data may otherwise be obtained from a database, a maps service, a satellite image database, a drone that is flown around the drive through location, etc.

(50) The process **400** includes defining a route of the drive through location (step **404**), according to some embodiments. In some embodiments, step **404** includes operating a user interface, display screen, computer screen, tablet, touch screen, etc., in order to display the map data obtained in step **402**. Step **404** may also include allowing the user, via a user interface, to provide one or more points that define the route. The points may be geographic (e.g., GPS) coordinates and may define different locations along a lane of traffic. In some embodiments, step **404** is performed by the setup device **302**. The setup device **302** may implement a curve-fitting technique (e.g., Catmull-Rom) based on the points defined by the user in order to determine the define the route. In some embodiments, step **404** includes defining multiple routes by allowing the user to input, via a user device, multiple points for each of multiple routes. For example, the different routes may correspond to different lanes of traffic (e.g., multiple order lanes, emergency lanes, parking lot locations or lanes of traffic, pickup lanes, entry lanes, exit lanes, etc.).

(51) The process **400** includes identifying locations for imaging devices along the route (step **406**), according to some embodiments. In some embodiments, step **406** is optional. For example, if cameras are already installed at the drive through location, then step **406** may be unnecessary. In some embodiments, step **406** is performed automatically by the setup device **302** based on an identification of the routes, possible mounting locations, as well as camera models, fields of view, etc. In some embodiments, step **406** is performed by a manufacturer or installation service of a vision system for the drive through location. For example, the map data and the route(s) obtained in steps **402-404** may be provided to the manufacturer or installation service. The manufacturer or installation service may provide suggestions regarding the locations, orientations, and models of

the imaging devices (e.g., cameras) in order to obtain image data of the entirety of the route from multiple locations.

(52) The process **400** includes installing imaging devices at the identified locations along the route (step **408**), according to some embodiments. In some embodiments, step **408** includes installing imaging devices (e.g., cameras) at the locations identified in step **406** about the drive through location. Step **408** may include installing posts or poles and imaging devices at the top of the posts or poles for locations where additional imaging devices are required. In some embodiments, step **408** is optional if the drive through location already has imaging devices installed on the premises. Step **408** may be performed by the manufacturer of the vision system for the drive through location, an installation service, etc.

(53) The process **400** includes defining zones along the route in image data obtained from the imaging devices (step **410**), according to some embodiments. In some embodiments, step **410** is performed by the setup device **302** based on inputs from a user (e.g., via the user interface **304**). For example, step **410** may include defining multiple zones (e.g., geometric shapes, areas, etc.) along the routes determined in step **404** for each of multiple imaging devices. In some embodiments, step **410** is performed by first prompting the user to define multiple zones or locations along the routes for the map data of the drive through location. In some embodiments, the setup device **302** is configured to automatically map the zones defined by the user (or automatically by defining zones along the routes) in the map data to the image data of the imaging devices. In some embodiments, the setup device **302** is configured to use one or more of a location, orientation, height, field of view, etc., of the imaging devices in order to automatically identify preliminary boundaries (e.g., edges, corners, etc.) for the zones in the image data of the imaging devices. The zones along the route may be definable by the user for the image data of the imaging devices and/or may be adjustable if an initial definition of the zones is provided automatically. The setup device **302** may prompt the user to define the zones in a manner that is consistent across the image data obtained from the imaging devices. For example, if a zone A is defined at a first order location (e.g., an order placement location in front of the drive through unit **108**) in image data obtained from a first imaging device, and the image data obtained from a second imaging device includes the first order location from a different perspective, the user may be prompted to define a zone at the first order location in the image data obtained from the second imaging device as zone A. In this way, the zones may be consistently defined in a uniform manner between the image data obtained from multiple imaging devices in order to account for different perspectives or views of a same location along the routes. Step **410** may be performed for any number of routes that are defined in step **404**.

(54) The process **400** includes defining locations of interest along the route (step **412**), according to some embodiments. In some embodiments, step **412** is performed by the setup device **302** by prompting the user to input, via the user interface **304**, one or more locations of interest. The locations of interest may include an entry location, an exit location, a location with a best view of an incoming vehicle for use in image analysis, an entry to a first order lane, an entry to a second order lane, an entry to a third order lane, an entry to a fourth order lane, a first, second, third, or fourth order location, an exit to any order lane, a first window location, a second window location, a pickup order location, etc. In some embodiments, step **412** is performed by labelling one or more of the zones defined in step **410** along the route as any of the locations. In this way, identification of a vehicle in one of the zones may result in identification that the vehicle is at a location of interest (e.g., that a vehicle has pulled up to a first order location). The locations of interest may be used in order to integrate the vision system with a POS unit. The locations of interest may also be used in order to identify performance parameters of the drive through location (e.g., service time, total life cycle time of a vehicle in the drive through location, etc.).

(55) The process **400** includes masking the image data obtained from the imaging devices (step **414**), according to some embodiments. In some embodiments, step **414** is performed by the setup

device **302** and the user interface **304**. For example, the setup device **302** may operate the user interface **304** to provide image data of the imaging device and allowing the user to define one or more zones or areas of the image data within which vehicles are unlikely to be detected. The user interface **304** may provide a pen tool, an eraser tool, a paintbrush tool, etc., for the user to define the zones or areas of the image data as the masks. The masks defined in step **414** generally define regions of the image data obtained from each of the imaging devices that should not be used or considered by the vision system when tracking vehicles through the drive through location. For example, if the image data from the imaging device includes surrounding areas (e.g., sidewalks, background image data, surroundings, etc.) where a vehicle is unlikely to be detected, the mask may remove the corresponding regions of the image data from consideration in image analysis and tracking. In some embodiments, the masks indicate areas of the image data that should be considered by the image analysis and tracking techniques (e.g., an area of interest). The masks can facilitate speed and accuracy of the image analysis and tracking techniques performed on the image data obtained from the imaging device. In some embodiments, the masks define areas or regions that have extraneous image data.

(56) The process **400** includes determining a profile for the drive through location (step **416**), according to some embodiments. In some embodiments, the profile is a configuration file or dataset for a controller (e.g., the controller **102**) of the drive through location. The profile may include the map data obtained in step **402**, the route(s) defined in step **402**, the locations and orientations of each of the imaging devices, the zones defined in step **410**, the locations of interest defined in step **412**, the masks defined in step **414**, and an image analysis technique. The profile may be location-specific for the particular drive through location and accounts for the unique camera placement, lanes of traffic shapes, distances, curvatures, the parking lot size, overall store arrangement, order locations, pickup locations, etc., of the specific drive through location for which process **400** is performed. In some embodiments, the profile is a locally-executable file for the controller of the drive through location. The profile may configure the controller of the drive through location to perform one or more image analysis and vehicle tracking techniques as desired. In some embodiments, the profile utilizes one or more of a neural network, a machine learning technique, an artificial intelligence tool, etc., that is provided by the manufacturer of the vision system described herein that can be implemented locally on the controller of the drive through location. If updates to the profile are desired, a drive through location administrator may re-initiate any of steps **402-416** in order to account for updates to lane configuration, construction at the store, closing of order lanes, etc. Step **416** may be performed by the setup device **302** by providing the profile to the controller **102**.

(57) The process **400** includes operating a POS unit of the drive through location using a vision system that implements the profile (step **418**), according to some embodiments. In some embodiments, step **418** is implemented by the controller **102** of the drive through location. For example, the controller **102** may implement the profile and identify incoming customers' vehicles and track the customers vehicles through the various lanes of traffic (e.g., along the routes). The controller **102** may track the vehicles as the vehicles travel along the routes and pass from one zone to the next along the routes. In response to the vehicle approaching to an order location, the controller **102** may operate a POS unit (e.g., associated with an order taker) to notify the order taker that a customer has arrived at the order location. The POS unit may be operated to prompt the order taker to initiate an order for the customer at the order location, and may include providing information regarding the customer at the order location (e.g., color of vehicle, type of vehicle, etc.). Step **418** may also include operating a display screen or an alert device (e.g., an audiovisual system, an alarm, an LED display, a speaker configured to provide an audible notification) to drive through personnel fulfilling orders to customers that pull up to an order pickup location. For example, step **418** may include operating the display screen to notify personnel regarding the POS data, as well as the type of the vehicle and the color of the vehicle as the vehicle is identified at the

order pickup location. Advantageously, notifying the personnel who fulfill orders to the customers regarding the POS data, as well as identifying characteristics of the vehicle associated with the POS data improves order fulfillment accuracy and reduces a likelihood that the wrong order is provided to the customers (e.g., reducing order mix-ups).

(58) Referring to FIG. 5, a diagram illustrates map data **500** (e.g., satellite data) of a particular restaurant **514**. The map data **500** includes a route **502** (e.g., a drive through route, a lane of travel, etc.) that extends around the restaurant **514** from an entry point to an order location, to a payment or pickup location, to an end point. In some embodiments, the route **502** is user-defined by presenting the map data **500** to the user (e.g., the owner of the restaurant **514**) and allowing the user to define points for the route **502**. In some embodiments, the map data **500** includes camera locations **504** that can be automatically suggested and populated on the map data **500** by the setup device **302**. The camera locations **504** may be provided as suggested locations by an administrator or manufacturer of a vision system. The camera locations **504** are provided such that cameras or imaging devices, when positioned at the camera locations **504**, are configured to obtain image data of an entirety of the route **502**.

(59) Referring to FIG. 6, a graphical user interface (GUI) **600** illustrates information that may be displayed on the user interface **304** during setup, according to some embodiments. In some embodiments, the GUI **600** includes an editing window **602a** (e.g., a first window). The editing window **602a** presents image data **610** from a selected one of the cameras **114**. For example, as shown in FIG. 6, the editing window **602a** includes image data **610** associated with “Camera 2.” The user may define a boundary **634** of a mask **608** in the editing window **602a**. In some embodiments, the user may define the boundary **634** of the mask **608** using a mouse cursor, a pen tool, a paint brush, a stylus on a tablet, etc. The mask **608** defines portions of the image data (e.g., ranges or areas of pixel) of the associated camera **114** that should be disregarded by an image analysis technique of the controller **102**. The mask **608** generally causes the controller **102** to disregard portions of the image data **610** that are surrounding or background image data of lanes of travel. In some embodiments, the mask **608** defines areas of interest and disinterest in the image data **610** obtained from the corresponding camera **114**. For example, the mask **608** may focus the controller **102** to analyze portions of the image data **610** corresponding to lanes of travel of the restaurant **14**.

(60) The GUI **600** also includes a toolbar **622** including a camera selector **624** (e.g., buttons, a drop-down menu, selectable options, etc.) and a lane selector **626**, according to some embodiments. In some embodiments, the lane selector **626** allows the user to select between multiple lanes to edit a series of zones **604** or zones **606** corresponding to the selected lane. In the example shown in FIG. 6, the restaurant **14** includes two lanes that are viewable by the “Camera 2.” The user may select between a first lane and a second lane in order to edit zones for the first lane or the second lane. In the example shown in FIG. 6, the first lane includes a series of zones **604**, shown as first zone **604a**, second zone **604b**, and third zone **604c**. The second lane includes a similar series of zones **706**, shown as first zone **704a**, second zone **704b**, and third zone **606c**. In some embodiments, when the second lane is selected at the lane selector **626**, the user may control points or corners of each of the zones **606** (e.g., adding points, moving points, removing points, etc.) in order to define the zones **606**. Once the user has appropriately defined the corners of the zones **606** of the second lane, the user may select the first lane via the lane selector **626** and repeat the process for the first lane. The user may define correspondingly labeled zones (e.g., first zone, second zone, third zone, etc.) for lanes of each of the cameras **114**. For example, the zone **604b** of the first lane is proximate a window in the image data **610** shown in the editing window **602a** (e.g., the “Camera 2” image data). The user may similarly define the second zone **604b** as in front of the window for image data obtained from other cameras **114** that also include the window such that the labeling of the zones **604** and **606** is consistent across different camera **114**.

(61) Referring still to FIG. 6, the GUI **600** includes a first preview pane **602b**, a second preview

pane **602c**, and a third preview pane **602d**. The second preview pane **602c** may correspond to go “Camera 2.” The first preview pane **602b** corresponds to “Camera 1” and the third preview pane **602c** corresponds to “Camera 3.” In some embodiments, in response to receiving a selection of different cameras **114** via the camera selector **624**, the GUI **600** provides the image data **610** of the selected camera **114** on the editing window **602a**. The user may then define the mask **608** and boundary **634** of the mask for each of the cameras **114**, as well as the zones **604** or **606** for the lanes, similarly as described above with reference to “Camera 2.”

(62) In some embodiments, the GUI **600** includes a first side pane **628** (e.g., an information display area) corresponding to the mask **608**. The first side pane **628** displays information corresponding to points or pixels of the boundary of the mask **634**. The GUI **600** includes a second side pane **630** (e.g., an information display area) corresponding to the first lane and the zones **604** of the first lane. In some embodiments, the second side pane **630** illustrates which pixels or points of the corresponding camera **114** define the zones **604** (of the first lane). In some embodiments, the GUI **600** includes a third side pane **632** that illustrates which pixels or points of the corresponding camera **114** define the zones **606** (of the second lane). In some embodiments, the information displayed in the first side pane **628**, the second side pane **630**, and the third side pane **632** are unique for each of the cameras **114**. The information displayed in the first side pane **628**, the second pane **630**, and the third side pane **632** may be a part of the setup profile that is provided from the setup device **302** to the controller **102** for each camera **114** at the restaurant **14**.

(63) Referring to FIG. 7, a diagram **700** illustrates the image data **610** of the “Camera 2” shown in FIG. 6 in an on-line implementation of the setup profile, according to some embodiments. As shown in FIG. 7, the image data **610** of the “Camera 2” includes masked out image data at mask **608** (e.g., clipped portions of the image data **610**, disregarded portions of the image data **610**, etc.). In some embodiments, the masked out image data, shown blocked by mask **608**, are disregarded by the controller **102** during implementation of image analysis techniques in order to improve the performance of the controller **102**.

(64) Referring to FIG. 8, various lanes of travel may be defined by the user or store owner about the store **14**. The lanes of travel can include, but are not limited to, a first order lane **802a**, a second order lane **802b**, a pickup lane **802c**, an emergency or exit lane **802d** (e.g., a fire lane), an entry lane **802e**, and a return lane **802f**. In some embodiments, the various lanes of travel include or are defined by paths **804** that extend through the lanes **802**. For example, the first order lane **802a** includes a first order lane path **804a**, the second order lane **802b** includes a second order lane path **804b**, the pickup lane **802c** includes a pickup path **804c**, the emergency or exit lane **802d** includes an exit path **804d**, the entry lane **802e** includes an entry path **804e**, and the return lane **802f** includes a return path **804f**. Each of the lanes **802** may include corresponding zones **810** defined along the paths **804** having unique identifiers. In some embodiments, the paths **804** and corresponding zones **810** are defined by the user during creation of the setup profile as described in greater detail above with reference to FIGS. 3-7. The paths **804** and zones **810** may be consistently defined between different perspectives of cameras **114** such that the cameras **114** provide a comprehensive and unified tracking of vehicles **30** along the zones **810** and paths **804**.

(65) The exit lane **802d** may be an escape or emergency lane to provide a point of egress from the pickup lane **802c**. The exit lane **802d** may be defined from a point of the pickup lane **802c** to a road or exit location surrounding the store **14**. In some embodiments, the return lane **802f** includes the return path **804f** that wraps around from the pickup lane **802c** to one or more parking spots **808** at a front of the store **14**. The parking spots **808** may each include corresponding zones for parking to await a carry out order. In some embodiments, the parking spots **808** are positioned at a pickup location **806** at a front of the store **14**.

(66) It should be understood that the vision system **200** that is setup for the store **14** as shown in FIG. 8 may include cameras **114** configured to obtain image data of all of the paths **804**. The cameras **114** may be configured to provide the image data of each of multiple zones **810** along the

paths **804** to the controller **102** during on-line implementation of the setup profile.

(67) Online Implementation

(68) Referring to FIG. **9**, the cloud computing system **104** may provide the setup profile to the controller **102** once the setup profile is fully generated. In some embodiments, the controller **102** and the vision system **200** are integrated into a POS system **900** of the drive through system **10**. The controller **102** is configured to use the setup profile in order to control operation of one or more POS units **902**. The controller **102** may also control operation of one or more alert devices **904** of the store **14** to provide feedback to employees of the store **14**. The controller **102** receives image data from the cameras **114** and uses the setup profile with the image data to track vehicles **30** through the drive through system **10**. In some embodiments, the controller **102** is configured to control operation of the POS units **902** based on the tracking and location of different vehicles **30** through the drive through system **10**.

(69) The POS units **902** may be positioned at any windows (e.g., windows **28**) of a restaurant or store. For example, both pickup windows at which orders are fulfilled and payment windows at which customers pay for their items may include corresponding POS units **902**, in addition to other display screens. In another example, pickup windows (e.g., an order fulfillment window) may include both a display screen (e.g., a kitchen display screen, a tablet, a fulfillment display screen, etc.) and a POS unit **902**. The display screens and the POS units **902** may be operated by the controller **102** based on the image data from the cameras **114** in order to provide characteristics of the vehicles **30** (e.g., type and color) according to one or more modalities (e.g., textual information, visual information such as icon that represent the characteristics of the vehicles **30**). The display screens and the POS units **902** may be operated based on tracking the vehicles **30** in a virtual space based on the zones that are defined when setting up the profile for the image data of each of the cameras **114**. As described in greater detail above, the zones may be defined differently for the image data of each camera **114** in order to account for different perspectives, orientations, positions, fields of view, etc., of the cameras **114** which results in the image data of the zones spanning different areas of the image data (e.g., different pixel regions). The definition of the zones accounts for the different locations, sizes, etc., of the image data of the zones in the image data of the cameras **114** such that the controller **102** can consistently track vehicles **30** through the drive through system **10** from camera to camera. The definition of the zones facilitates seamless tracking of the vehicles **30** through the drive through system **10** despite different perspectives and positioning of the camera **114**.

(70) The controller **102** may obtain the image data from each of the cameras **114** in real-time. In some embodiments, when a vehicle **30** first arrives at the drive through system **10**, the controller **102** may use an initial image of the vehicle **30** to determine (1) a color of the vehicle **30**, and (2) a type of the vehicle **30**. In some embodiments, the controller **102** is configured to implement an image analysis technique using the initial image of the vehicle **30** to determine (1) the color of the vehicle **30**, and (2) the type of the vehicle **30**. In some embodiments, the controller **102** includes different predetermined colors such as red, blue, orange, black, white, green, etc. In some embodiments, the controller **102** is configured to determine, using the image analysis technique and the initial image of the vehicle **30**, which of the predetermined colors to which the color of the vehicle **30** corresponds. In some embodiments, the controller **102** includes different predetermined types of vehicles such as car, truck, SUV, Jeep, etc. In some embodiments, the controller **102** is configured to determine, using the image analysis technique and the initial image of the vehicle **30**, which of the predetermined types of vehicle to which the type of the vehicle **30** corresponds.

(71) In some embodiments, the controller **102** is configured to use the paths and zones along the paths in the setup file to track location of the vehicle **30** throughout a lifecycle of the vehicle **30**. In some embodiments, the controller **102** is configured to track which of the zones the vehicle **30** travels along or at which the vehicle **30** is currently located. For example, the controller **102** may identify that the vehicle **30** is traveling along the first order lane path **804a** by detecting the vehicle

30 at zones along the first order lane path **804a**. In some embodiments, the controller **102** is configured to use the image analysis techniques and one or more locations of interest (e.g., tagged or identified zones) to determine if the vehicle **30** has arrived at a specific location. The controller **102** may use the locations of interest (e.g., the tagged or identified zones) and the detection of the vehicles **30** at the locations of interest to operate corresponding POS units **902**, or the alert devices **904**. For example, in response to detecting that the vehicle **30** has arrived at a first order location, the controller **102** may operate corresponding POS units **902** to notify staff (e.g., order taking staff) that the vehicle **30** is awaiting service at a particular drive through unit **108**. The controller **102** may operate the POS units **902** to notify the order taking staff regarding the color and the type of the vehicle **30** that is at the particular drive through unit **108**. In this way, the vision system **200** (e.g., the image data of the cameras **114**, the image analysis and tracking techniques performed by the controller **102**, etc.) can be integrated with the POS units **902** of the store **14**. The POS units **902** may be display screens, the order taking stations **110**, etc.

(72) In some embodiments, the POS units **902** may be operated by the order taking staff. POS data resulting from the POS units **902** may be provided to the controller **102**. The controller **102** may operate alert devices **904** responsive to at least one of the POS data obtained from the POS units **902**, or based on results of the image analysis techniques performed using the image data.

(73) In some embodiments, the controller **102** is configured to operate a pickup alert device **908** or a carry-out alert device **910**. In some embodiments, the pickup alert device **908** is a display screen positioned proximate a window at which the vehicle **30** pulls up or arrives at in order to receive their order. The controller **102** may track the vehicle **30** to the window, and in response to the vehicle **30** arriving at the window, operate the pickup alert device **908** to display the color of the vehicle **30** and the type of the vehicle **30** that is at the window. In some embodiments, the controller **102** is also configured to operate the pickup alert device **908** to display POS data such as an order number, items in the order, a name of the customer in the vehicle **30**, etc. In this way, the POS data of the customer in the vehicle **30** may be tied to the detected color and type of the vehicle **30**, which may be displayed to staff as the staff fulfill the order to the customer in the vehicle **30**.

Advantageously, displaying POS data (e.g., order information) in combination with characteristics of the vehicle **30** to receive the order (e.g., the type and color of the vehicle **30**) facilitates accurate order fulfillment and reduces a likelihood of the incorrect order or items being handed to customers.

(74) Referring still to FIG. 9, the controller **102** is also configured to operate the carry-out alert device **910** to notify carry out staff regarding a customer awaiting carry out. In some embodiments, for example, during busy times of the day, the staff may instruct customers in the vehicles **30** to pull around to the pickup location **806**. The controller **102** may receive an input from the POS units **902** that a specific customer has been asked to pull to the pickup location **806**. The controller **102** is configured to track the vehicle **30** of the specific customer as the vehicle **30** travels to the pickup location **806**, and identify a zone at which the vehicle **30** parks (e.g., which of the parking spots **808** that the vehicle **30** parks at). The controller **102** may operate the carry-out alert device **910** to notify the carry out staff regarding the POS data (e.g., the order data, a customer's first name, etc.), the characteristics of the customer's vehicle **30** (e.g., the color and type of the car), as well as which of the parking spots **808** that the vehicle **30** has parked at. In some embodiments, the carry-out alert device **910** is a display screen (e.g., a touch screen, a user interface, etc.). The carry-out alert device **910** may provide the POS data, the characteristics of the customer's vehicle **30**, and which of the parking spots **808** the vehicle **30** is located by presenting textual information, a graphical representation of the vehicle **30** and the corresponding parking spot **808**, or any combination thereof.

(75) Referring still to FIG. 9, the controller **102** may also operate the drive through unit(s) **108** and one or more audio communications devices **912**. In some embodiments, the audio communications devices are wearable headphones or wearable communications devices configured to be worn by staff or personnel of the store **14** in order to establish communications with the drive through units

108 (e.g., audio communications). The audio communications devices **912** may each include one or more speakers and one or more microphones for providing and obtaining audio (e.g., the microphones are configured to obtain audio input, and the speakers are configured to provide audio output). The drive through units **108** may similarly include speakers and microphones for providing a second end of audio communications. In some embodiments, the controller **102** is configured to implement one or more audio channels to facilitate bi-directional audio communication between the audio communications devices **912** and the drive through units **108**. In some embodiments, the audio communications device **912** are operated based on the vehicle **30** detected at an order location by the controller **102** to prompt the order taker to switch to a channel to take the customer's order.

(76) The controller **102** may also be configured to count a number of customers (e.g., a number of vehicles **30**) that are in line awaiting order. In some embodiments, the number of customers that are in line at the drive through system **10** awaiting their turn to place an order is referred to as a “stack size.” In some embodiments, during busy times of day, the controller **102** may identify that a number of vehicles **30** have wrapped around the store **14**, into a parking lot, onto the street, etc. The controller **102** may use the image data provided by the cameras **114** of surrounding areas of the store **14** (e.g., the parking lot, an adjacent road, an entrance to the parking lot, etc.), and determine the stack size of the drive through system **10**. In some embodiments, the controller **102** is configured to record a time of day and corresponding stack size. The time of day and corresponding stack size may be used by the controller **102** in order to track busy times of day and initiate preparatory actions for preparing food and/or beverage items before the busy times of day.

(77) In some embodiments, the controller **102** is configured to operate one or more kitchen alert devices **906** responsive to the stack size. In response to detecting a stack size above a threshold using the image data provided by the cameras **114**, the controller **102** may initiate cooking operations by prompting kitchen staff to prepare food and/or beverage items. The controller **102** may initiate the cooking operations or preparatory actions (e.g., a kitchen action) by operating the one or more kitchen alert devices **906**. In some embodiments, the controller **102** uses a predetermined set of instructions based on the stack size to determine which preparatory actions to initiate. For example, if the controller **102** includes historical data that 90% of customers typically order a specific type of food and/or beverage item that requires a significant amount of time to prepare, the controller **102** may prompt the kitchen staff, based on the stack size, to initiate preparation of a corresponding amount of the food and/or beverage item. In this way, the controller **102** may initiate kitchen options using the image data provided by the cameras **114**. In some embodiments, the controller **102** is configured to operate the POS unit(s) **902** or a manager display screen **914** to display the stack size of the drive through system **10**.

(78) Referring to FIG. **10**, the POS system **900** is shown in greater detail. The controller **102** is shown to include includes processing circuitry **1002**, a processor **1004**, and memory **1006**. Processing circuitry **1002** can be communicably connected to a communications interface such that processing circuitry **1002** and the various components thereof can send and receive data via the communications interface. Processor **1004** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

(79) Memory **1006** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **1006** can be or include volatile memory or non-volatile memory. Memory **1006** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, memory **1006** is communicably connected to processor **1004** via processing circuitry **1002** and includes computer

code for executing (e.g., by processing circuitry **1002** and/or processor **1004**) one or more processes described herein.

(80) In some embodiments, controller **102** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments controller **102** can be distributed across multiple servers or computers (e.g., that can exist in distributed locations). For example, controller **102** can be made up of multiple controllers or microcontrollers that individually or collectively are configured to carry out operations as described herein.

(81) The controller **102** is configured to obtain the setup profile from the cloud computing system **104**, and store the setup profile in the memory **1006**, shown as setup profile **1020**. The controller **102** may use the setup profile in order to implement the techniques described herein, or to configure the controller **102** to implement the techniques as described herein. In some embodiments, the controller **102** is also configured to provide POS data and performance metrics to the cloud computing system **104**. The controller **102** may also receive POS data or performance metrics of other drive through systems **10** from the cloud computing system **104**.

(82) The controller **102** includes a database **1008**, an order taking manager **1010**, a communications manager **1012**, and a vision system manager **1030**, according to some embodiments. The database **1008** may be configured to store POS data, menu data, performance data, historical metrics of busy times, inventory or stock, personnel identification, etc. In some embodiments, the order taking manager **1010** is configured to retrieve menu data from the database **1008** for use in constructing a GUI of the POS units **902**, or to update the GUI of the POS units **902** (e.g., when an item is out of stock, when a price of the item has changed, etc.). In some embodiments, the order taking manager **1010** is configured to implement any of the techniques as described in U.S. application Ser. No. 17/325,871, now U.S. Pat. No. 11,244,681, filed May 20, 2021, the entire disclosure of which is incorporated by reference herein.

(83) Referring still to FIG. **10**, the communications manager **1012** is configured to route traffic between the drive through units **108** and the wearable communications device **112**. In particular, the communications manager **1012** may define various channels such that the wearable communications devices **112** are configured to switch between the various channels. Each of the drive through units **108** may communicate on an assigned channel that the wearable communications device **112** can switch between. In particular, the communications manager **1012** is configured to route audio traffic from the drive through units **108** (e.g., drive through audio) to the wearable communications devices **112** and vice versa (e.g., the communications provided by the wearable communications devices **112** are provided to the drive through units **108**). In some embodiments, the switching of the wearable communications devices **112** is achieved by selection or input on the display screen of the POS units **902**.

(84) The vision system manager **1030** is configured to integrate the vision system **200** (e.g., the cameras **114**) with the POS units **902**. The vision system manager **1030** may be configured to implement an image analysis technique **1014** using the image data obtained from the cameras **114**. In some embodiments, the image analysis technique **1014** is implemented using a neural network, a machine learning technique, a deep learning technique, an artificial intelligence, etc. The image analysis **1014** is configured to detect the presence of a vehicle in the image data provided by the cameras **114** as well as a color and type of the vehicle.

(85) The vision system manager **1030** also includes a vehicle tracker **1028**. The vehicle tracker **1028** is configured to use the setup profile **1020** (e.g., the zones of the setup profile **1020**) and track the vehicles detected by the image analysis **1014** through the drive through system **10**. In particular, the vehicle tracker **1028** may use the detection of the color and type of the vehicle **30** provided by the image analysis **1014** in combination with a map including the zones. The vehicle tracker **1028** may output notifications of the vehicle **30** traveling into zones of interest. The vehicle tracker **1028** is configured to track position, speed, change in position, etc., of the vehicles **30** in the drive through system **10**.

(86) The GUI manager **1016** is configured to use the tracking of the vehicles **30** provided by the vehicle tracker **1028** and the color and type of the vehicles **30** provided by the image analysis **1014** to generate or adjust the GUI that is provided to the POS units **902**. In particular, when a vehicle arrives at one of the drive through units **108** as identified by the vehicle tracker **1028**, the GUI manager **1016** may update a corresponding icon of the GUI that is displayed on the POS units **902** to visually indicate the color and type of the vehicle **30**. The GUI manager **1016** may also adjust the GUI of the POS units **902** such that the POS units **902** display textual information of the color and type of the vehicle **30** that is present at the drive through units **108**. The order takers (e.g., users of the POS units **902**) may select corresponding drive through units **108** via the POS units **902** in order to switch their wearable communications device **112** to the drive through unit **108** at which the vehicle **30** is detected.

(87) In some embodiments, the order takers may initiate an order taking process for the customer of the vehicle **30** (e.g., by adding items to the order). Once the order taking process is complete, the POS unit **902** may provide the POS data (e.g., order information, order number, etc.) which is provided to the database **1008** and the vehicle tracker **1028**. The vehicle tracker **1028** may record an identifier of the POS data and tie the POS data with the vehicle **30**. In this way, as the image analysis **1014** continues to use newly obtained image data to identify the vehicle **30** in the image data, and the vehicle tracker **1028** tracks the vehicle **30** through the drive through system, the POS data (e.g., order data, customer name, items in order, total cost, etc.) may be linked with the vehicle **30**. Responsive to the vehicle tracker **1028** indicating that the vehicle **30** has arrived at a pay window or a pickup window (e.g., the first window **28a**, the second window **28b**, etc.), the GUI manager **1016** may operate POS units **902** to indicate that the vehicle **30** (with the linked POS data) has arrived at the pay window or the pickup window. Advantageously, the POS units **902** are integrated with the vision system **200** such that the POS units **902** can be updated responsive to tracking of vehicles **30** through the drive through system **10**. Similarly, the vehicle tracker **1028** may track vehicles through the drive through system **10** while linking the POS data of the vehicle **30** with the tracking. The GUIs that are generated by the GUI manager **1016** based on results of the vision system manager **1030** using the image data may be any of the GUIs described in greater detail below with reference to FIGS. 12-15.

(88) In some embodiments, the GUI manager **1016** is also configured to generate a manager GUI and operate the manager display **914** to provide the manager GUI. The manager GUI may be any of the GUIs described in greater detail with reference to FIGS. 16-18. In some embodiments, the manager display **914** is a separate display screen that is presented to all personnel of the store **14** on a separate display screen. In some embodiments, the manager display **914** is a store manager or administrator's display screen that is accessible remotely (e.g., via the cloud computing system **104**). The manager GUI generally displays an overall map of the drive through system **10**, icons of detected vehicles along the map of the drive through system **10** that visually indicate colors and types of the vehicles, and POS data for each of the vehicles **30** (e.g., an indication).

(89) In some embodiments, the POS data includes order data that may be selected and viewed by selecting the icons of the manager GUI. In some embodiments, the manager GUI is generated by the GUI manager **1016** using results of the vehicle tracker **1028**, the image analysis **1014**, the POS data, and the setup profile **1020**. The manager GUI may include a visualization (e.g., textual, an icon, etc.) of a stack size of vehicles **30** in the drive through system **10**. The stack size indicates a number of vehicles **30** that are beyond a certain point (e.g., upstream of the drive through units **108**, upstream of an entrance of the drive through system **10**, etc.) or between other vehicles **30** in the drive through system **10**. The manager GUI may also display various performance metrics or characteristics of the drive through system **10**. The manager GUI may also display vehicles **30** at a carry-out (take-out, pick-up) location (e.g., that have been ordered by staff to pull to a parking spot) and vehicles awaiting a mobile pickup order.

(90) The vision system manager **1030** includes a performance manager **1026** that is configured to

use results of the vehicle tracker **1028** to determine one or more performance characteristics of the drive through system **10**. In some embodiments, the performance manager **1026** is configured to identify, based on the results of the vehicle tracker **1028**, when a vehicle arrives at an entry point of the drive through system **10**. The performance manager **1026** may record a time at which each vehicle arrives at the entry point. The time at which the vehicle **30** arrives at the entry point may be linked to the particular vehicle **30**. The performance manager **1026** may also record a time at which the vehicles **30** each arrive at one of the drive through units **108**. The performance manager **1026** may also record a time at which the customers in the vehicles **30** begin their order (e.g., once the order taker switches to communicating via the drive through units **108**). The performance manager **1026** can also record a time at which the customers in the vehicles **30** complete their order. The performance manager **1026** may also record a time at which the customers in the vehicles **30** arrive at a first window (e.g., a payment window) and a second window (e.g., a pickup window). The performance manager **1026** may also record a time at which the customers in the vehicles **30** have their orders fulfilled. In some embodiments, the performance manager **1026** is configured to use any of the recorded times to estimate various quantities of elapsed time. For example, the performance manager **1026** may estimate a total amount of time that it takes each vehicle **30** to enter the drive through system **10**, place their order, pick up their order, and exit the drive through system **10**. The performance manager **1026** may determine an average amount of time it takes a vehicle to arrive, order, receive their order, and exit the drive through system **10**. In some embodiments, the performance manager **1026** is configured to track any of a number of vehicles that are commanded to pull over to a pickup location, a number of vehicles **30** that leave the drive through system **10** without ordering (e.g., due to the line being too long), an average amount of time that a customer waits at the drive through units **108** to begin placing an order, an average amount of time it takes a customer to place their order once initiated, number of vehicles **30** in the drive through system **10**, a number of customers served so far for the day, etc.

(91) The performance manager **1026** may be configured to provide any of the performance characteristics or metrics described herein to the cloud computing system **104**. The performance manager **1026** may receive performance metrics or rankings of the store **14** from the cloud computing system **104**. For example, the ranking may indicate which place the store **14** is in relative to other stores in the area or in a chain. The ranking of the store **14** may be determined based on any of the performance metrics or characteristics described herein (e.g., average time for a customer to enter the drive through system **10**, place their order, receive their order, and leave the drive through system **10**).

(92) In some embodiments, the performance manager **1026** is configured to provide any of the performance characteristics or metrics of the drive through system **10** to the GUI manager **1016** for display on the POS units **902** or the manager display **914**. In some embodiments, the performance manager **1026** is configured to store any of the performance metrics or characteristics of the drive through system **10** in the database **1008**. The performance manager **1026** may also provide ranking of the drive through system **10** to the GUI manager **1016** for display on the POS units **902** or the manager display **914**.

(93) The vision system manager **1030** may include a prediction engine **1024**, an action manager **1022**, and an alert manager **1018**, according to some embodiments. In some embodiments, the prediction engine **1024** is configured to use historical (e.g., seasonal historical data) of demand at the drive through **10** to predict actions that should be taken by kitchen staff or personnel of the drive through system **10**. For example, the prediction engine **1024** may predict busy times, and provide the results of the predicted busy times of the store **14** to the action manager **1022**. The action manager **1022** is configured to use the results of the prediction engine **1024** in order to identify actions that should be taken to prepare for busy times of the store **14**, and prompt the alert manager **1018**. The alert manager **1018** is configured to provide alerts to the alert devices **904** to notify staff of the store **14** regarding the identified actions. Identified actions may include

preparation of certain food and/or beverage items, an amount of food and/or beverage items to be prepared, sending out line busters with handheld POS units, opening new lanes, etc.

(94) The action manager **1022** may use outputs from the prediction engine **1024** and may also use outputs from the performance manager **1026**. In some embodiments, outputs of the performance manager **1026** include the stack size, and indicate real-time or current data of the drive through system **10** as opposed to predicted data. In some embodiments, the action manager **1022** may defer to using results of the performance manager **1026** which indicate real-time performance metrics as opposed to using the outputs of the prediction engine **1024**. For example, if the prediction engine **1024** predicts, based on historical data, that there is a likelihood that ten customers are present in the drive through system **10** at a current time, but the image data indicates that twenty-five customers are present in the drive through system **10** (e.g., as indicated by the outputs of the performance manager **1026**), the action manager **1022** may use the results of the image data instead of the outputs of the prediction engine **1024**. In some embodiments, the action manager **1022** is configured to use a machine learning technique in order to determine one or more cooking or kitchen operations that should be performed responsive to number of customers in the drive through system **10**. For example, the action manager **1022** may use a known percentage of customers that will order a specific type of food and/or beverage item, and initiate preparation of a corresponding amount of the specific type of food and/or beverage item based on the number of customers in the drive through system **10**.

(95) The action manager **1022** may also determine if it is appropriate to open an additional lane based on the current number of customers. In some embodiments, the action manager **1022** is configured to observe the average amount of time for a customer to enter the drive through system **10**, place their order, receive their order, and exit. In response to the average amount of time exceeding a threshold, the action manager **1022** may determine that the additional lane should be opened, and may either initiate the opening of the additional lane automatically, or may prompt a store manager to open the additional lane. Similarly, the action manager **1022** may initiate or prompt opening the additional lane in response to the total number of customers in the drive through system **10** exceeding a threshold or in response to the stack size exceeding a threshold.

(96) The alert manager **1018** is configured to receive actions or determinations of the action manager **1022**, and operate the alert devices **904** to prompt staff to perform the action(s) determined by the action manager **1022**. For example, the alert devices **904** may be kitchen alert devices such that the alert manager **1018** initiates preparation of food and/or beverage items in accordance with the determinations of the action manager **1022**. Similarly, the alert devices **904** may be alert devices for order takers or staff personnel to open new lanes. In some embodiments, the GUI manager **1016** is configured to operate the manager GUI to prompt the manager to initiate opening the additional lane. The manager of the store may be presented with a request to open the additional lane, and provide via the manager display **914** confirmation to open the additional lane. Responsive to receiving the confirmation from the manager, the alert manager **1018** may operate the alert devices **904** to notify order taking personnel to open the additional lane.

(97) Referring to FIG. **11**, a flow diagram of a process **1100** of operating a drive through system includes steps **1102-1108**, according to some embodiments. The process **1100** may be implemented by the controller **102** using local techniques without uploading image data to a cloud computing system. In some embodiments, the process **1100** illustrates implementation and use of the profile in process **400**. In some embodiments, the process **1100** as described herein is performed as step **418** of process **400** (e.g., implementing the profile).

(98) The process **1100** includes providing a drive through system including a vision system having a camera and a controller (step **1102**), according to some embodiments. The vision system may be installed at a drive through of a store or restaurant. The vision system can include the camera configured to obtain image data of different lanes, pickup windows, order locations, exit lanes, emergency lanes, parking lots, etc. In some embodiments, the vision system includes multiple

cameras that are positioned along different lanes of travel. The cameras may be positioned at locations, heights, and orientations as suggested or determined in step **406** of process **400**.

(99) The process **1100** includes detecting a vehicle using the image data (step **1104**), according to some embodiments. In some embodiments, step **1104** includes obtaining the image data at the controller from the camera. Detecting the vehicle may include identifying a color (e.g., black, red, blue, purple, green, brown, white, etc.) and a type of the vehicle (e.g., compact car, sedan, SUV, truck, etc.). In some embodiments, step **1104** includes using an image analysis technique (e.g., a machine learning technique, an artificial intelligence, a neural network, etc.) in order to predict the color and type of the vehicle in the image data. Step **1104** can also be used to detect a location of the vehicle. In some embodiments, step **1104** includes detecting which of multiple zones along various routes of the drive through system at which the detected vehicle is currently located. In some embodiments, step **1104** is performed simultaneously with steps **1106** and **1108** in real-time in order to track the vehicle's location. Step **1104** may also include detecting a speed or motion of the vehicle. Step **1104** may also include detecting if the vehicle is stopped or parked. In some embodiments step **1104** is performed by the controller (e.g., controller **102**).

(100) The process **1100** includes operating a POS unit based on the detection of the vehicle (step **1106**), according to some embodiments. In some embodiments, step **1106** includes operating the POS unit of an order taker in order to notify the order taker regarding the detection of the vehicle. For example, the POS unit may be operated to notify the order taker that the vehicle has arrived at an order location. In some embodiments, operating the POS unit includes prompting the order taker to initiate an order process. In some embodiments, operating the POS unit includes providing a notification of the color and type of the vehicle. In some embodiments, operating the POS unit includes providing a graphical representation of the vehicle that visually illustrates both the color and type of the vehicle. Step **1106** may include notifying an order taker to switch to an audio channel to take the customer's order. Step **1106** may be performed by the controller **102**.

(101) The process **1100** includes operating a display screen based on the detection of the vehicle (step **1108**), according to some embodiments. The display screen may be a display screen positioned within a kitchen of the store or restaurant, a display screen for a store owner or manager, a display screen at a station (e.g., an order take out station), etc. The alert may include providing a graphical representation of all vehicles in the drive through system. The alert may additionally or alternatively include recommendations to open a new lane (e.g., a store manager operation), perform cooking operations (e.g., a kitchen operation), carry out an order to a customer that is parked in a pickup zone (e.g., a take out operation), etc. The alert may additionally or alternatively include providing notifications regarding the color and type of the detected vehicle at a particular location. In some embodiments, the alert includes POS or order data provided simultaneously with at least one of a snapshot of the detected vehicle or the color and type of the detected vehicle. For example, the display screen may be positioned at an order pickup window facing staff that has the responsibility of handing orders (e.g., food and/or beverage items) to customers at the order pickup window. Step **1108** may include operating the display screen to provide both (1) order data (e.g., POS data, order number, order identification, first name of customer, etc.) and (2) vehicle data (e.g., a snapshot of the vehicle, the color and type of the vehicle, etc.). Providing the order data and the vehicle data simultaneously can facilitate improved order fulfillment accuracy and reduce a likelihood that one customer's order is handed to another customer.

(102) POS GUIs

(103) Referring to FIGS. **12-15**, the POS unit(s) **902** may be configured to display various graphical user interfaces (GUIs) provided by the controller **102**. The controller **102** may operate the POS units **902** to display the GUIs using the image data provided by the cameras **114**.

Advantageously, the controller **102** is configured to integrate the vision system **200** (e.g., the image data and the analysis performed by the controller **102** based on the image data) with the POS units **902**.

(104) Referring to FIG. 12, a first GUI **1200** includes a lane pane **1202** (e.g., a lane window), a windows pane **1204** (e.g., a windows lane), a carry out pane **1206**, and an order take out pane **1208**, according to some embodiments. In some embodiments, the lane pane **1202** is configured to display data regarding each of multiple lanes. In particular, the lane pane **1202** includes an entry, box, window, icon, widget, or pane corresponding to each drive through unit **108** of different lanes. The windows pane **1204** may provide data regarding customers that are present at multiple pickup windows of the drive through system **10**. The carry out pane **1206** may provide data regarding carry out orders (e.g., customers parked in parking spots **808**). The order take out pane **1208** may provide data regarding various take out orders.

(105) The lane pane **1202** may be configured to display data corresponding to each of the drive through units **108** and the corresponding vehicles **30** that are present at each of the drive through units **108** (either currently placing an order, or awaiting to place an order). In particular, the lane pane **1202** may display both the color and the type of the car that is present at the drive through units **108** as determined by the controller **102** using the image data. For example, as shown in FIG. 12, a blue SUV is currently present at the drive through unit **108** of lane **3**, and a black car is present at the drive through unit **108** of lane **1**. The lane pane **1202** may be updated in real-time by the controller **102** based on determinations by the controller **102** using the image data. For example, when the controller **102** determines, using the image data, that a blue SUV has arrived at the drive through unit **108** of lane **3**, the controller **102** may adjust the GUI **1200** of the POS units **902** in order to notify the order takers that a blue SUV has arrived at the drive through unit **108** of lane **3**. In some embodiments, the lane pane **1202** includes multiple selectable icons that display the color and type of the vehicles **30** that are present at the drive through units **108** of each lane. In some embodiments, GUI **1200** is configured to display vehicle information relating to the characteristic of the vehicle and order information for an order received from an occupant of a drive through to provide compiled order information. The vehicle information and order information may be processed at a single one of the POS units **902** or by multiple POS units **902**. Further, controller **102** is configured to update a display of the compiled order information during a service journey of the vehicle at the drive through. The service journey corresponds to the total course from entry to exit at the drive through.

(106) Referring to FIGS. 13-14, the GUI **1200** may be configured to open a new window or widget **1218** responsive to selection of one of the icons of the windows pane **1202**. In some embodiments, the widget **1218** includes a check in button **1210** (e.g., icon) and a new order button **1212**. For example, the order taker may select the check in button **1210** in order to initiate checking in a customer that has placed an online or mobile order and has arrived at the drive through system **10** for pickup. The order taker may select the new order button **1212** in order to initiate a POS and add items to the customer's order. In some embodiments, the widget **1218** includes vehicle information **1214**, and an option **1216** to change the vehicle information **1214**. In some embodiments, the vehicle information **1214** is populated by the controller **102** using the detected characteristics of the vehicle **30** (e.g., the color and type of the vehicle). Once the POS has been completed and the order has been placed, the POS data (e.g., the order data) may be tied to the corresponding vehicle information **1214** throughout the rest of the vehicle **30**'s lifecycle through the drive through system **10**.

(107) The GUI **1200** may similarly be provided to fulfillment staff via the pickup alert device **908** (e.g., a display screen, a touch screen, etc.). However, the GUI **1200** as provided to the fulfillment staff may exclude the options for adding items to the customer's order. The GUI **1200** may present both the type and color of the vehicle **30** that is currently present at the corresponding window (e.g., the red truck at window **1**) along with order information that is tagged to the characteristics of the vehicle **30** (e.g., the type and color).

(108) Referring to FIG. 15, the windows pane **1204** may include multiple selectable icons corresponding to multiple pickup windows of the store **14**. The selectable icons (e.g., widgets,

boxes, buttons, etc.) may include vehicle information, similar to the lane pane **1202** indicating both the color and type of vehicle **30** that is currently present at the windows of the store **14**. The vehicle information of the windows pane **1204** may be populated or updated by the controller **102** based on the determinations obtained by the controller **102** using the image data. When the vehicle **30** arrives at the window (e.g., “red truck” shown at “Window **1**” in FIG. **15**), the staff at the corresponding window (e.g., Window **1**) may select the corresponding icon. Responsive to selection of the corresponding icon, the GUI **1200** displays an order pane **1220**. The order pane **1220** can include vehicle information **1222** illustrating the color and type of the vehicle **30** that is present at the window. The order pane **1220** can also include a check in button **1224** and an add order button **1226**, similar to as shown in FIGS. **13-14**. The order pane **1220** can also include order information **1228** and a serve button **1230**. In this way, the GUI **1200** is configured to display the order information **1228** in combination with the vehicle information **1222**. Advantageously, providing both the order information **1228** in combination with the vehicle information **1222** facilitates reduced likelihood of incorrect order fulfillment.

(109) Referring again to FIGS. **12-15**, the lane pane **1202** and the windows pane **1204** may provide textual information of the vehicle data (e.g., the color and type of vehicle) as well as graphical information of the vehicle data. For example, FIG. **12** illustrates an icon or graphical illustration of a black car, a blue SUV, and a red truck.

(110) Tracking GUI

(111) Referring to FIGS. **16-18**, the controller **102** may operate a display screen (e.g., the manager display **914**) to include a GUI that illustrates live positions of all vehicles at the drive through system **10**. The GUI may be an overall or universal GUI that is determined by the controller **102** based on the image data of the cameras **114**.

(112) Referring particularly to FIG. **16**, the controller **102** may generate and operate the manager display **914** to provide a GUI **1600**. The GUI **1600** may include a graphical representation of the store **14**, shown as store icon **1602**. The GUI **1600** may also include graphical representations of vehicles **30** in the drive through system **10**, shown as vehicle icons **1618**. In some embodiments, the vehicle icons **1618** are selected in order to visually indicate both the color and type of the vehicle as detected by the controller **102**. The vehicle icons **1618** are located and adjusted in their location in real-time corresponding to live detection of the vehicles **30** in the drive through system **10**.

(113) The GUI **1600** may include menu icons **1604**, corresponding to the drive through units **108**. In particular, the GUI **1600** may include menu icon **1604a** and menu icon **1604b** corresponding to two different drive through units **108**. The GUI **1600** illustrates which vehicle is currently located at the drive through units **108**. The GUI **1600** may also include a pay window icon **1606** and a pickup window icon **1608**. The GUI **1600** may visually indicate which of the vehicles **30** are at the pay window **1606** and the pickup window **1608**. The GUI **1600** can also include a finish line **1612** that visually indicates which customers are leaving the drive through system **10** after a successful order. In some embodiments, the GUI **1600** also visually illustrates vehicles **30** that have been requested to pull to a pickup location, shown as pickup spots **1614**. The GUI **1600** may place vehicle icons **1618** corresponding to vehicles **30** that are currently detected at the pickup spots **1614**. In some embodiments, the GUI **1600** also visually illustrates vehicles **30** that are awaiting mobile order pickup, shown as mobile order pickup spots **1616**.

(114) Referring still to FIG. **16**, the GUI **1600** may also include stack size indicators, shown as stack size icon **1620**. The stack size icon **1620** visually illustrates a number of customers that are present between two points, or beyond a certain point. As shown in FIG. **16**, the stack size icon **1620** illustrates that seven vehicles are present between the pay window and the pickup window, visually illustrated by pay window icon **1606** and pickup window icon **1608**.

(115) Referring to FIG. **17**, the controller **102** may generate and operate the manager display **914** to provide a GUI **1700**. The GUI **1700** may be similar to the GUI **1600**. The GUI **1700** may include

vehicle icons **1718**, menu icons **1704**, a finish line **1712**, a pay window icon **1606**, and a pickup window icon **1708**. Similarly to GUI **1600**, the GUI **1700** can include pickup spots **1714** and mobile order pickup spots **1716**. The GUI **1700** also includes a stack size **1720** that indicates a number of customers that are beyond a certain point (e.g., vehicles **30** that are in line for the drive through system **10** upstream of the point at which the stack size **1720** is displayed).

(116) Referring to FIG. **18**, the GUI **1700** is shown in a simplified view, illustrating only one lane of traffic of the drive through system **10**. The GUI **1700** may also present a drive through performance tab **1740** illustrating various performance characteristics of the drive through system **10**. In particular, the drive through performance tab **1740** may include a car count **1742** (e.g., a number of vehicles that are currently in the drive through system **10**), a total car count **1744** (e.g., a total number of vehicles or customers that have been tracked through the drive through system **10** for the day so far), an average menu time **1746** (e.g., an average amount of time that a customer is at the drive through unit **108**), and a number of lost customers **1748** (e.g., a number of customers that leave the drive through system **10** due to long lines). The GUI **1700** may also include a performance dashboard **1750** (e.g., a window, a pane, an area, a report, etc.) that includes a performance icon **1752** indicating overall performance of the drive through system **10**. In particular, the performance dashboard **1750** may indicate an average amount of time **1754** for a customer to enter, order, pay, receive their order, and exit the drive through system **10**. In some embodiments, the performance dashboard **1750** includes a leaderboard **1756** that indicates a ranking of the drive through system **10** of the store **14** that is illustrated by the GUI **1700**, relative to other drive through locations. The ranking of the leaderboard **1756** may be determined based on the average amount of time **1754**.

(117) Point of Sale and Kitchen GUIs

(118) Referring to FIGS. **19-31**, the controller **102** may operate any of the POS units (e.g., POS units **902**) to display the GUIs **1800** shown in FIGS. **19-31** at least partially using the image data obtained from the cameras **114**. In particular, the GUIs can display a screen for a director of a drive-through system to view, for example, a total number of mobile orders, and identify the source of particular mobile orders, e.g., from different services. For each mobile order, a user, a processor, a unique order identifier, a time of ordering, and a total amount can be displayed. The GUIs can also display, simultaneously with the order information, a lane drop-down menu and a window drop-down menu, allowing for selection of a particular lane and/or window of interest to a user to be selected to display additional information. As shown in FIGS. **19-31**, the POS units can be operated to provide GUIs in real-time that are updated based on identified visual characteristics of the vehicles **30** in the drive through system **10** and updated as the vehicles **30** are tracked through the drive through system **10**. For example, as shown in FIG. **21** a blue truck arrives at lane **3**, an order is placed, and the GUI is updated when the blue truck pulls around to the second window as shown in FIG. **31**. Accordingly, the GUI can display, for example, a detected vehicle, a check-in button indicating the customer in the detected vehicle has checked in, and a corresponding order. In addition, mobile order information can also be displayed.

(119) Referring particularly to FIG. **19**, the GUI **1800** includes a lanes pane **1802**, a windows pane **1804**, a mobile orders pane **1806**, and one or more delivery service panes, shown as delivery service pane **1808** and delivery service pane **1810** (e.g., third party delivery services). When a vehicle arrives at the drive through unit **108** at the second lane (e.g., a white SUV), the GUI **1800** is updated in real-time as shown in FIG. **20**. In some embodiments, all of the POS units **902** that display the GUI **1800** are updated in real-time responsive to detection of vehicles in the drive through **10**. As shown in FIG. **21**, another vehicle, a blue truck, has arrived at the third lane. An icon of the vehicle may indicate in real-time whether another order taker has switched to communicating (e.g., via audio communications) with the vehicle and is taking the customer's order. As shown in FIG. **22**, when the user of the POS unit **902** on which the GUI **1800** is displayed selects the blue truck icon at the third lane, the GUI **1800** presents various options **1812**. The

options **1812** include a button **1818** to change a characteristic of the detected vehicle (e.g., if the vehicle that is detected by the vision system **200** has a different characteristic than what is identified by the vision system **200**). For example, if the vehicle detected is not a blue truck and is instead a green SUV, the user may select the button **1818** to correct the characteristics detected by the vision system **200**.

(120) Referring still to FIG. **22**, the options **1812** also include a check-in button **1814**, and a new order button **1816**. The user may select the check-in button **1814** and select an appropriate mobile order from the mobile order pane **1806** or one of the delivery panes **1808** and **1810** if the customer in the blue truck is picking up an order that has already been placed (e.g., by a mobile application or for delivery). If the customer in the blue truck is a new customer that has not yet placed an order, the user may select the new order button **1816**.

(121) Responsive to selecting the new order button **1816**, the user (e.g., the order taker) is presented with a menu, shown in FIG. **23**. The order taker may select items from the menu in order to add to the customer's order. As shown in FIG. **23**, the order menu may include a selectable switch **1824** (e.g., a toggle button) to select between breakfast, lunch, and dinner menus. The order menu may also include a ribbon of item options **1822** including burgers, drinks, sides, meals, combos, etc. Selection of the different items options **1822** causes the menu to display different menu items that can be added to the customer's order. As shown in FIG. **23**, the combos menu option is selected, and the menu displays combo items **1826** and limited time offers **1828**. The order taker may select these items and add them to the customer's order as instructed by the customer. The order taker may also select between a carry out option and a dine in option for the new order.

(122) Referring to FIG. **24**, responsive to selecting items to add to the customer's order, the menu displays a current order **1820** including items that has been added to the order, and a cost associated with each item. The current order **1820** may mirror or reflect an order that is displayed to the customer via the drive through unit **108** in real-time. Once the customer's order is completed, the order taker may select a pay or complete order button.

(123) Referring to FIG. **25**, once the customer's order has been completed, the associated icon for the blue truck includes an indicator bar (e.g., a green indicator bar) to show that the customer has completed an order. The indicator may be provided on all the POS units **902** that display the GUI **1800** so that other order takers are informed that the customer in the blue truck has already placed their order and does not require further assistance at that lane. At this point, the order taker may, for example, select the icon for the green car as shown in FIGS. **26** and **27** or the white SUV as shown in FIG. **25** to start the next order.

(124) Referring to FIGS. **28** and **29**, the white SUV which was previously in lane **2** (shown in FIG. **5**) has now pulled around to the third window, shown in the windows pane **1804** in order to place an order at the window in order to pay. As shown in FIG. **31**, the blue truck has pulled up to the second window in order to pay and pickup their order. The flow of traffic from the drive through units **108** to the pickup windows is updated in real-time across all the POS units **902** based on tracking of the vehicles by the vision system **200**. When a vehicle (e.g., the white SUV) pulls up to the third window, the windows pane **1804** indicates the presence of the white SUV by providing a textual representation (e.g., "White SUV") and a graphical representation (e.g., an icon showing a white SUV). Responsive to selection of the pane or icon corresponding to the white SUV, the GUI **1800** presents the options **1812** to the user. The GUI **1800** also presents an order pane **1820** which indicates an order number, a total amount of time the vehicle has been in the drive through **10**, and a required payment amount.

(125) Responsive to the blue truck arriving at the second window, shown in FIG. **31**, the GUI **1800** updates in real-time to show the user which window the blue truck is currently at. The user may select the blue truck icon to view the order information of the customer in the blue truck, similarly to FIG. **30** for the white SUV. In some embodiments, the GUI **1800** shown in FIGS. **25-31** illustrate

a director screen for a manager or director of the restaurant to oversee drive through activity and details. In some embodiments, the GUI **1800** shown in FIGS. **19-24** illustrate a screen that is presented to an order taker or cashier of the drive through **10**.

(126) Referring to FIG. **32**, a kitchen GUI **1900** illustrates various orders that may be presented to kitchen staff (e.g., an employee of the drive through **10** preparing orders). The operation of the kitchen GUI **1900** is coordinated with order placement at the POS units **902** that implement the GUI **1800** described in detail above with reference to FIGS. **19-31**. The controller **102** may be configured to operate the kitchen display screen (e.g., a display device, the kitchen alert devices **906**, a tablet, etc.) to display the kitchen GUI shown in FIG. **32**. As shown in FIGS. **32**, the kitchen display screen is operated in real-time to include newly placed orders and associated visual characteristics of the vehicles **30** for the orders. The kitchen GUI **1900** can be updated in real-time based on operations at the POS units (e.g., updated to include items that are added to the order). The kitchen screen GUIs indicate both the visual characteristics of the vehicles **30** (e.g., “blue truck”) as well as order information that the customers in the vehicles **30** have placed. When an order is fulfilled or the items have been prepared, the controller **102** may operate the kitchen display screen(s) to remove corresponding widgets, panes, or windows (e.g., as shown in FIGS. **37** and **38** where the grey car order window is removed).

(127) As shown in FIG. **32**, the kitchen GUI **1900** includes various widgets **1902-1908** that are populated in real-time responsive to order placement. Each of the widgets **1902-1908** corresponds to a different order and lists the various items in the order. For example, the widget **1902** illustrates a burger meal and a chicken sandwich, as well as a lane that the vehicle is in (e.g., the white SUV), an identification of the vehicle (e.g., “white SUV”), an order number, whether the vehicle is in the drive through or not, and whether the order has been paid for or not. Likewise, the widget **1904** is populated for the blue truck responsive to order placement at the POS unit **902**, the widget **1906** is populated for the green car, and the widget **1908** is populated for the grey car. In some embodiments, the widgets **1902-1908** are updated in real-time based on feedback from the vision system **200**. For example, when the white SUV arrives at the third window (FIG. **28**), the widget **1902** may be updated to read “WINDOW 2” instead of “LANE 2.”

(128) Referring to FIGS. **12-15**, **16-18**, and **19-32**, the controller **102** may be configured to operate the POS units **902**, the kitchen alert devices **906** (e.g., kitchen display screens), and other display screens (e.g., the manager display **914**) to provide various GUIs in a coordinated manner based on the image data and tracking performed based on the image data. As shown in FIGS. **16-18**, the manager display **914** may be operated provide a tracking GUI that illustrates visual indications of the locations of vehicles **30** in the drive through system **10**. As the vehicles **30** travel through the drive through system **10** (that is, as they transit through the drive through on their service journey), the tracking GUI may be updated in real-time to reflect current location and amount elapsed time since the vehicles **30** have entered the drive through system **10** (e.g., indicated by the status ring with increasing shaded region surrounding the icons of the vehicles **30**). When the vehicles **30** arrive at the drive through or order units **108**, the controller **102** may update the GUIs of the POS units **902** (shown in FIGS. **12-15** and **19-31**) to indicate that the vehicles **30** have arrived at the drive through units **108** as well as visual characteristics of the vehicles **30**. As the customers in the vehicles **30** place their orders and the order takers enter order information via the POS units **902**, the kitchen GUI **1900** (shown in FIG. **32**) may be operated by the controller **102** to indicate the items added to the order and corresponding visual characteristics of the vehicle **30** of the customer (e.g., type and color of vehicle). When the vehicles **30** arrive at pickup windows, the POS GUI (GUI **1800**) may be updated to notify order takers or order fulfillment personnel regarding the vehicle **30** at the windows as well as the visual characteristics of the vehicle **30** at the window.

(129) It should be understood that any of the GUIs described herein with reference to FIGS. **19-32** may be implemented by the controller **102** and provided on a variety of display screens throughout the drive through **10** in a coordinated manner. The GUIs described herein with reference to FIGS.

19-32 may be the result of detecting and tracking vehicles 30 through the drive through system 10 in order to provide real-time feedback or prompts to kitchen staff, order taking staff, fulfillment staff, restaurant managers, analysis systems, etc.

Configuration of Exemplary Embodiments

(130) As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(131) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(132) The term “coupled,” as used herein, means the joining of two members directly or indirectly to one another. Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. Such members may be coupled mechanically and/or electrically.

(133) The term “or,” as used herein, is used in its inclusive sense (and not in its exclusive sense) so that when used to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is understood to convey that an element may be either X, Y, Z; X and Y; X and Z; Y and Z; or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

(134) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” etc.) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(135) The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

(136) In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device, etc.) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile

memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit and/or the processor) the one or more processes described herein. References to “a” or “the” processor should be understood to encompass a plurality of processors individually or collectively configured to carry out operations as described herein.

(137) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(138) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(139) It is important to note that the arrangement of the system as shown in the various exemplary embodiments is illustrative only. All such modifications are intended to be included within the scope of the present disclosure. Other substitutions, modifications, changes, and omissions may be made in the design and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. A drive through system comprising: a point of sale unit configured to be operated to implement a point of sale; a plurality of cameras configured to obtain image data of a lane of the drive through system; and processing circuitry configured to: determine, using the image data, a characteristic of a vehicle at the lane waiting to place an order; operate a display screen of the point of sale unit to provide a textual or visual description of the characteristic of the vehicle waiting to place the order to prompt an order taker to communicate with a customer in the vehicle and obtain order data for the customer; execute a point of sale transaction at the point of sale unit; and responsive to detection of one or more of a plurality of vehicles using the image data, perform at least one of (i) initiate an alert via the point of sale unit for staff of the drive through system, (ii) initiate an action to open an additional lane, or (iii) initiate a kitchen action to prepare one or more food or beverage

items.

2. The drive through system of claim 1, wherein the characteristic of the vehicle comprises a type of the vehicle selected from a plurality of predetermined types of vehicles using the image data, and a color of the vehicle.
3. The drive through system of claim 2, wherein the plurality of predetermined types of vehicles comprise at least a car, a truck, and a sport utility vehicle.
4. The drive through system of claim 1, wherein the processing circuitry is further configured to: determine, using the image data from the plurality of cameras and a plurality of predetermined zones, a location of the vehicle along a path of the lane, the plurality of predetermined zones defined for the image data of each of the plurality of cameras to account for different orientations and positions of the plurality of cameras to produce image data of the plurality of predetermined zones at different portions of the image data of each of the plurality of cameras; and responsive to the vehicle arriving at an order placement location of the plurality of predetermined zones, operate the point of sale unit to provide the textual or visual description of the characteristic of the vehicle.
5. The drive through system of claim 1, wherein the textual or visual description of the characteristic of the vehicle comprises a graphical representation of the vehicle.
6. The drive through system of claim 1, wherein the drive through system further comprises at least one of a point of sale unit or a screen visible to staff at a pickup window of the drive through system, the processing circuitry configured to operate at least one of the point of sale unit or the screen at the pickup window to provide order data associated with the vehicle and the characteristic of the vehicle responsive to the vehicle arriving at the pickup window.
7. The drive through system of claim 1, wherein the processing circuitry is configured to: track the plurality of vehicles through the lane of the drive through system using the image data obtained from the plurality of cameras; and operate, based on the image data, a carry-out alert device to notify carry-out staff regarding (i) the characteristic of a vehicle tracked to a carry-out location, (ii) an identification of a carry-out location where the vehicle is positioned among a plurality of carry-out locations, and (iii) order data for the vehicle at the carry-out location.
8. A method of controlling a drive through system, the method comprising: determining, using image data of a lane of the drive through system obtained from a camera, a characteristic of a vehicle at the lane waiting to place an order; operating a display screen of a point of sale unit to provide a textual or visual description of the characteristic of the vehicle waiting to place the order to prompt an order taker to communicate with a customer in the vehicle and obtain order data for the customer; executing a point of sale transaction at the point of sale unit; and responsive to detection of one or more of a plurality of vehicles using the image data, performing at least one of (i) initiating an alert via the point of sale unit for staff of the drive through system, (ii) initiating an action to open an additional lane, or (iii) initiating a kitchen action to prepare one or more food or beverage items.
9. The method of claim 8, wherein the characteristic of the vehicle comprises a type of the vehicle selected from a plurality of predetermined types of vehicles using the image data, and a color of the vehicle.
10. The method of claim 9, wherein the plurality of predetermined types of vehicles comprise at least a car, a truck, and a sport utility vehicle.
11. The method of claim 8, further comprising: determining, using the image data from a plurality of cameras and a plurality of predetermined zones, a location of the vehicle along a path of the lane, the plurality of predetermined zones defined for the image data of each of the plurality of cameras to account for different orientations and positions of the plurality of cameras to produce image data of the plurality of predetermined zones at different portions of the image data of each of the plurality of cameras; and responsive to the vehicle arriving at an order placement location of the plurality of predetermined zones, operating the point of sale unit to provide the textual or visual description of the characteristic of the vehicle.

12. The method of claim 8, wherein the textual or visual description of the characteristic of the vehicle comprises a graphical representation of the vehicle.
 13. The method of claim 8, further comprising operating at least one of a point of sale unit or a screen at a pickup window to provide order data associated with the vehicle and the characteristic of the vehicle responsive to the vehicle arriving at the pickup window, the screen visible to staff at the pickup window of the drive through system.
 14. The method of claim 8, further comprising: tracking the plurality of vehicles through the lane of the drive through system using image data obtained from a plurality of cameras; and operating, based on the image data, a carry-out alert device to notify carry-out staff regarding (i) the characteristic of a vehicle tracked to a carry-out location, (ii) an identification of a carry-out location where the vehicle is positioned among a plurality of carry-out locations, and (iii) order data for the vehicle at the carry-out location.
 15. A non-transitory computer readable medium configured to store instructions that, when executed by at least one processor, cause the at least one processor to perform operations comprising: determining, using image data of a lane of a drive through system obtained from a camera, a characteristic of a vehicle at the lane waiting to place an order; operating a display screen of a point of sale unit to provide a textual or visual description of the characteristic of the vehicle waiting to place the order to prompt an order taker to communicate with a customer in the vehicle and obtain order data for the customer; executing a point of sale transaction at the point of sale unit; and responsive to detection of one or more of a plurality of vehicles using the image data, performing at least one of (i) initiating an alert via the point of sale unit for staff of the drive through system, (ii) initiating an action to open an additional lane, or (iii) initiating a kitchen action to prepare one or more food or beverage items.
 16. The non-transitory computer readable medium of claim 15, wherein the characteristic of the vehicle comprises a type of the vehicle selected from a plurality of predetermined types of vehicles using the image data, and a color of the vehicle, wherein the plurality of predetermined types of vehicles comprise at least a car, a truck, and a sport utility vehicle.
 17. The non-transitory computer readable medium of claim 15, the operations further comprising: determining, using the image data from a plurality of cameras and a plurality of predetermined zones, a location of the vehicle along a path of the lane, the plurality of predetermined zones defined for the image data of each of the plurality of cameras to account for different orientations and positions of the plurality of cameras to produce image data of the plurality of predetermined zones at different portions of the image data of each of the plurality of cameras; and responsive to the vehicle arriving at an order placement location of the plurality of predetermined zones, operating the point of sale unit to provide the textual or visual description of the characteristic of the vehicle.
 18. The non-transitory computer readable medium of claim 15, wherein the textual or visual description of the characteristic of the vehicle comprises a graphical representation of the vehicle.
 19. The non-transitory computer readable medium of claim 15, the operations further comprising operating at least one of a point of sale unit or a screen at a pickup window to provide order data associated with the vehicle and the characteristic of the vehicle responsive to the vehicle arriving at the pickup window, the screen visible to staff at the pickup window of the drive through system.
 20. The non-transitory computer readable medium of claim 15, the operations further comprising: tracking the plurality of vehicles through the lane of the drive through system using image data obtained from a plurality of cameras; and operating, based on the image data, a carry-out alert device to notify carry-out staff regarding (i) the characteristic of a vehicle tracked to a carry-out location, (ii) an identification of a carry-out location where the vehicle is positioned among a plurality of carry-out locations, and (iii) order data for the vehicle at the carry-out location.
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